

11.1 Direct-address tables

Direct addressing is a simple technique that works well when the universe U of keys is reasonably small. Suppose that an application needs a dynamic set in which each element has a key drawn from the universe $U = \{0, 1, \dots, m-1\}$, where m is not too large. We shall assume that no two elements have the same key.

To represent the dynamic set, we use an array, or *direct-address table*, denoted by $T[0..m-1]$, in which each position, or *slot*, corresponds to a key in the universe U . Figure 11.1 illustrates the approach; slot k points to an element in the set with key k . If the set contains no element with key k , then $T[k] = \text{NIL}$.

The dictionary operations are trivial to implement:

DIRECT-ADDRESS-SEARCH(T, k)

1 **return** $T[k]$

DIRECT-ADDRESS-INSERT(T, x)

1 $T[x.\text{key}] = x$

DIRECT-ADDRESS-DELETE(T, x)

1 $T[x.\text{key}] = \text{NIL}$

Each of these operations takes only $O(1)$ time.

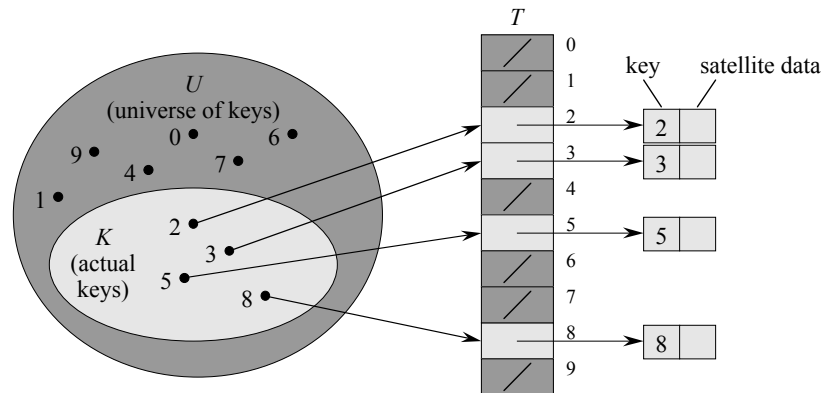


Figure 11.1 How to implement a dynamic set by a direct-address table T . Each key in the universe $U = \{0, 1, \dots, 9\}$ corresponds to an index in the table. The set $K = \{2, 3, 5, 8\}$ of actual keys determines the slots in the table that contain pointers to elements. The other slots, heavily shaded, contain NIL.